FIRSTEDGE BANK

EXECUTIVE BUSINESS INTELLIGENCE REPORT

CUSTOMER CHURN RISK AND RETENTION STRATEGY ANALYSIS

An executive briefing highlighting customer attrition trends, financial risk, and strategic retention opportunities.

Prepared for

FirstEdge Bank Executive Management

Prepared by

Adekunle Balikis

Data & Business Analyst

June 2026

Tools: Power BI • Power Query • DAX

Contents

Customer Churn and Attrition Analysis

Executive Summary

Problem Classification: Diagnostic + Strategic | Primary Value Driver: Revenue Preservation & Customer Retention

Project Overview

FirstEdge Bank is experiencing elevated customer attrition, creating significant financial and operational challenges across its retail banking portfolio. Customer churn not only reduces long-term customer lifetime value but also increases acquisition costs, weakens customer loyalty, and exposes the bank to substantial revenue risk through lost deposits and reduced product utilization. This analysis was conducted to identify the primary drivers of customer attrition, quantify its business impact, and provide actionable recommendations that support data-driven retention strategies

Business Problem Definition

FirstEdge Bank is facing a significant customer retention challenge within its retail banking portfolio, with Approximately 51.7% of the bank's retail customer base has churned either closed their accounts or become inactive. This trend directly affects the bank's ability to sustain long-term revenue growth, strengthen customer loyalty, and maximize customer lifetime value. Executive stakeholders—including the Chief Executive Officer, Chief Risk Officer, Head of Retail Banking, and Customer Retention team—require a clearer understanding of which customers are most at risk, why they are leaving, and where intervention efforts should be prioritized.

Customer attrition is not a one-time event but a continuous business risk that compounds over time. Although year-over-year churn has declined by approximately **50%**, indicating encouraging progress, maintaining and improving this trend requires proactive action. The highest levels of churn are concentrated in the **East (55.9%)** and **North (53.2%)** regions, suggesting that regional performance, customer engagement, pricing strategy, and onboarding processes require closer evaluation to prevent further customer losses.

The financial implications are substantial. Approximately **₦3 billion** in customer deposits are associated with churned accounts, representing a direct revenue and balance sheet risk. As customer acquisition is considerably more expensive than customer retention, continued attrition—particularly among customers within their first two years of banking—threatens the sustainability of the bank's growth strategy and increases the cost of replacing lost customers.

This analysis was undertaken to identify the customer segments most vulnerable to churn, quantify the financial exposure associated with customer attrition, and uncover the key behavioral and operational factors influencing customer departures. Success will be measured by reducing the overall churn rate from **51.7% to below 35% within the next 12 months**, decreasing inactivity among customers aged **46–55**, increasing product adoption during customers' first two years with the bank, and reducing churn in the East and North regions by at least **10 percentage points**.

Analysis Objectives

The primary objective of this analysis is to provide FirstEdge Bank's executive leadership with actionable insights into the drivers, financial impact, and distribution of customer churn. Specifically, the analysis aims to:

- Quantify the overall level of customer attrition and its impact on revenue exposure.
- Identify the customer demographics, behaviors, and regions with the highest churn risk.
- Evaluate the relationship between customer engagement, product ownership, and retention.
- Determine the primary reasons customers leave the bank and assess their business implications.
- Provide evidence-based recommendations that support targeted retention strategies and sustainable business growth.

Key Findings

1,000	**51.7%**	**₦3bn**	**−50%**
Customers Analyzed	Churn Rate	Deposits at Risk	YoY Churn Change

Strategic Insight

1. **Customer attrition is driven primarily by competitive pressure, not operational failure.**

Better Competitor is the leading reason customers leave the bank, ahead of high fees, poor service, and financial difficulty. Churned customers recorded a complaint level broadly comparable to retained customers (complaint gap: **−0.11**), suggesting that service complaints alone do not explain customer attrition.

Business implication the bank's greatest retention challenge is maintaining a competitive value proposition rather than addressing isolated service failures. Retention strategies focused solely on customer service improvements are unlikely to produce meaningful reductions in churn.

2. **Inactivity is a leading indicator of churn (the 46–55 age segment is the highest-risk zone)**

Inactive customers churn at a measurably higher rate than active ones (53.13% active churn rate, with the 46–55 age band recording the highest overall churn at 56.28%). The risk matrix analysis shows that within this age group, inactive customers represent an acute concentration of risk. This is a predictable, detectable signal the bank currently has no mechanism to act on it before the customer leaves.

Business implication the bank should implement an inactivity early-warning flag: accounts dormant beyond a defined threshold (e.g., 60 days with no transaction) should automatically trigger a proactive retention outreach. Catching customers before formal churn is possible, but only with a system to detect and act on inactivity signals in real time

3. **New customers churn faster, and cross-selling is underused in the critical first two years.**

Customers in the 'New' tenure band (0–2 years) churn at 54.29%, the highest of any tenure group. Separately, customers with only one banking product show higher churn rates than those with two or more. These two findings connect: the bank is failing to deepen relationships quickly enough during the period when customers are most likely to leave. Onboarding is the weakest link in the customer lifecycle.

Business implication Every additional product a customer holds increases their switching cost and ties them more deeply to the bank. A structured 90-day onboarding journey - with a targeted prompt to adopt a second product - is one of the highest-ROI retention investments the bank can make right now.

4. **Customer churn represents a material financial exposure, with approximately ₦3 billion in deposits associated with churned accounts.**

₦3 billion in deposits linked to churned customers; East and North regions have the highest churn rates.

Business implication Retention is not only a customer experience issue but also a balance sheet and revenue protection priority. Prioritize high-value customers in the highest-risk regions with targeted retention programs and ongoing executive monitoring.

Recommendations and Action Plan

The following actions are prioritized by urgency and expected business impact.

1. **Launch a competitive pricing and product (review within 30 days).** 'Better Competitor' is the #1 churn driver. Commission a structured comparison of FirstEdge Bank's savings rates, fixed deposit yields, and fee structures against the top three competitor banks in the East and North regions, where churn is highest. Without understanding the competitive gap, retention efforts are blind.
2. **Deploy an inactivity early-warning system (within 60 days).** Define an inactivity threshold (recommended: 60 days without a transaction or login) and route flagged accounts to a dedicated retention outreach team. Prioritize customers aged 46–55 and those in the East and North regions. This converts a lagging indicator (churn) into a leading one (inactivity), giving the bank a window to intervene.
3. **Redesign the new customer onboarding journey (within 90 days).** Introduce a structured 90-day onboarding programme that includes a targeted prompt for customers to adopt a second banking product by day 45. Customers with multiple products churn at a lower rate; onboarding is the highest-leverage point to build that stickiness early.
4. **Introduce relationship-managed retention for high-balance customers (within 60 days).** Analysis shows that high-credit-score, high-balance customers churn at rates similar to the overall base — meaning the bank's most financially valuable customers are not receiving differentiated retention treatment. Assign a relationship manager touchpoint to this segment with quarterly check-ins and preferential product access.
5. **Rebalance the retention budget away from complaint-handling and toward competitive response.** The data shows complaint volume does not significantly differ between customers who stay and customers who leave. Complaint resolution is necessary but not sufficient. Retention investment should be reallocated proportionally toward the pricing, onboarding, and inactivity interventions above, which have stronger evidence of impact.

Success Metrics

The success of the recommended retention strategy should be evaluated through a defined set of performance indicators. These metrics provide executive management with measurable targets to monitor progress, assess the effectiveness of retention initiatives, and support ongoing strategic decision-making.

Metric	Current Baseline	12-Month Target
Overall Churn Rate	51.7%	Below 35%
East Region Churn Rate	55.9%	Below 45%
North Region Churn Rate	53.2%	Below 43%
Avg Products per New Customer (Yr 1)	~1.0	1.5+
Churn rate among inactive customer (Age 46–55)	56.28%	Below 45%

Supporting Documentation

The business findings presented in this report are supported by a three-page interactive Power BI dashboard comprising:

- *Executive Dashboard**:** Provides a high-level overview of customer churn, financial exposure, and key performance indicators.*
- *Customer Demographics Analysis**:** Examines churn across demographic, regional, and behavioral segments to identify high-risk customer groups.*
- *Financial and Risk Analysis: Assesses financial exposure, customer value, and risk patterns through analytical visualizations, including a credit score versus account balance scatter plot and an age-by-activity risk matrix.*

Supporting technical documentation includes:

- *DAX Measures Documentation*
- *Data Model (Star Schema) Documentation*

*Data Note: This analysis is based on a synthetically generated dataset designed to reflect common banking data structures and real-world data quality challenges. The observed churn rate of **51.7%** is intentionally higher than typical industry benchmarks (approximately 15–25%) to ensure sufficient analytical variation across customer segments. While the numerical results should not be interpreted as representative of a specific financial institution, the analytical methodology, business insights, and recommendations are applicable to real-world customer retention scenarios.*

DAX Measures

This provides a complete reference for all 18 DAX measures built in the FirstEdge Bank Customer Churn Analysis project. Measures are organized into five functional folders matching the display folder structure inside Power BI. Each entry includes the full DAX formula, the analytical purpose of the measure, how it is used in the report, and the formatting applied.

1. **Core Metrics**

   Core metrics provide the primary KPIs used to evaluate customer attrition, portfolio size, retention performance, and customer engagement. These measures form the foundation of the executive dashboard.

Measure	**Purpose**	**Business Use**	**DAX Formula**
Total Customer	Counts the total number of unique customer rows in Dim_Customer. Serves as the baseline denominator for all churn rate and percentage calculations across the report.	Displayed on the Executive Summary page as the headline customer count KPI. Referenced inside Churn Rate and % of Total Balance at Risk.	COUNTROWS(Dim_Customer)
Churned Customer	Counts customers whose Churn_Status is recorded as 'Yes'. Uses CALCULATE to apply a filter context over the churn_status fact table, isolating only churned records from the full row count.	Core churn volume metric used across all three report pages. Serves as the base measure referenced inside all time intelligence calculations (YTD,	Churned Customer = CALCULATE(COUNTROWS(churn_status), churn_status[Churn_Status] = "Yes")

Churn Rate	Calculates the proportion of customers who have churned as a percentage of the total customer base. DIVIDE is used instead of the division operator to handle zero denominators gracefully and return 0 rather than an error.	Primary performance indicator for executive leadership and the retention team. Displayed prominently on Pages 1 and 2. Used as the base measure inside High Net Worth Churn Rate, Active Churn Rate, and Inactive Churn Rate.	Churn Rate = DIVIDE([Churned Customer], [Total Customer], 0)
Retention Rate	The arithmetic complement of Churn Rate — the proportion of customers who remained. Derived directly from the Churn Rate measure to ensure both metrics always sum to 100%.	Provides executives the positive framing alongside the churn rate figure — both sides of the same coin displayed together on the Executive Summary KPI strip.	Retention Rate = 1 - [Churn Rate]
Average Account Balance	Calculates the mean account balance across all customers in the current filter context. Responds to slicers and cross-filtering from other visuals	Used to understand the financial profile of the customer base and provide context for Revenue at Risk figures. Displayed as a supporting KPI on the Demographics page.	Average Account Balance = AVERAGE(Dim_Customer[Account_Balance])
Total Complaints	Calculates total recorded customer complaints.	Displayed on the Risk Analysis page as a headline operational metric. Provides the raw complaint volume that feeds into the Avg Complaints — Churned and Avg Complaints — Retained measures.	Total Complaints = SUM(churn_status[Num_Of_Complaints])

2. **Time Intelligence**

Time Intelligence measures monitor churn performance over time, enabling management to identify trends, compare historical periods, and evaluate whether retention initiatives are improving outcomes.

Measure	Purpose	Business Use	DAX Formula
Churned Customer YTD	Calculates the cumulative count of churned customers from the start of the calendar year to the latest available date in the current filter context. TOTALYTD is a time intelligence function that resets at the start of each year.	Allows leadership to track whether churn is accelerating or decelerating as the year progresses. Displayed on the Executive Summary page paired against Churned Customer Prior Year for direct comparison.	Churned Customer YTD = TOTALYTD([Churned Customer], Dates[Date])
Churned Customer Prior Year	Returns churn for the equivalent period in the previous year.	Paired with Churned Customer YTD on the Executive Summary page to show whether the bank's churn trajectory is improving or worsening year-over-year.	Churned Customer Prior Year = CALCULATE([Churned Customer], SAMEPERIODLASTYEAR(Dates[Date]))
Churned Growth %	Calculates percentage change in churn versus the previous year.	The −50% YoY improvement displayed on the Executive Summary page is derived from this measure. Provides leadership with a single directional metric for whether churn is getting better or worse.	Churned Growth % = DIVIDE([Churned Customer] - [Churned Customer Prior Year], [Churned Customer Prior Year], 0)
3-Month Moving Avg Churn	Calculates a rolling 3-month average of churned customers by iterating over the three months ending at the latest date in context	Allows leadership to distinguish genuine trend changes from seasonal spikes or data	3-Month Moving Avg Churn = AVERAGEX(DATESINPERIOD(Dates[Date], LASTDATE(Dates[Date]), -3,

	using AVERAGEX and DATESINPERIOD. Smooth out short-term volatility to reveal the underlying churn trend direction.	anomalies. A rising moving average signals an accelerating churn problem.	MONTH), [Churned Customer])

3. **Financial Impact**

These measures quantify the financial exposure associated with customer churn. They estimate the value of deposits at risk and measure the proportion of the bank's portfolio affected by customer attrition, helping management understand the potential impact on revenue and liquidity.

Measure	Purpose	Business Use	DAX Formula
Revenue at Risk	Calculates the total account balances associated with churned customers, representing deposits at risk of loss.	The ₦3 billion figure displayed on Pages 1 and 3 is derived from this measure. It is the single most executive-facing metric in the report and is referenced in the % of Total Balance at Risk calculation.	Revenue at Risk = CALCULATE(SUM(Dim_Customer[Account_Balance]), churn_status[Churn_Status] = "Yes")
% of Total Balance at Risk	Calculates the proportion of total customer balances exposed through churned accounts	The ₦3 billion figure displayed on Pages 1 and 3 is derived from this measure. It is the single most executive-facing metric in the report and is referenced in the % of Total Balance at Risk calculation.	% of Total Balance at Risk = DIVIDE([Revenue at Risk], SUM(Dim_Customer[Account_Balance]), 0)

4. **Operational Risk**

These measures assess whether operational factors, particularly customer complaints, are associated with churn behaviour. By comparing complaint patterns between churned and retained customers, they help determine whether operational performance is a significant driver of customer attrition.

Measure	Purpose	Business Use	DAX Formula
Avg Complaints – Churned	Calculates the average number of complaints lodged by churned customers	Tests whether high-complaint customers are more likely to leave. Displayed as a supporting metric on the Risk Analysis page.	Avg Complaints - Churned = CALCULATE(AVERAGE(churn_status[Num_Of_Complaints]), churn_status[Churn_Status] = "Yes")
Avg Complaints – Retained	Calculates the average number of complaints submitted by retained customers.	Used to determine whether service quality is a meaningful differentiator between churned and retained customers.	Avg Complaints - Retained = CALCULATE(AVERAGE(churn_status[Num_Of_Complaints]), churn_status[Churn_Status] = "No")
Complaint Gap	Calculates the difference in average complaints between churned and retained customers. . A positive value indicates churned customers complained more. A negative or near-zero value indicates complaints are not a meaningful predictor of churn.	The near-zero result (−0.11) directly challenges the assumption that service improvement alone will reduce churn, redirecting leadership attention toward competitive pricing instead.	Complaint Gap = [Avg Complaints - Churned] - [Avg Complaints - Retained]

5. **Risk Metrics**

These measures identify customer segments with elevated likelihood of churn based on behavioral and financial characteristics**.** They are used in customer segmentation analyses, risk matrices, and targeted retention assessments**.**

Measures	Purpose	Business Use	DAX Formula
High Net Worth Churn Rate	Filters the Churn Rate measure to customers in the High balance tier only — those with account balances above ₦3,000,000. Uses CALCULATE to override the filter context and isolate this specific segment.	Displayed on the Risk Analysis page. If High Net Worth Churn Rate exceeds the overall Churn Rate, it signals the bank is losing its most financially valuable customers at a disproportionate rate.	High Net Worth Churn Rate = CALCULATE([Churn Rate], Dim_Customer[Balance_Tier] = "High")
Active Churn Rate	Calculates the churn rate specifically among customers currently classified as active. Provides the comparison baseline for the Inactive Churn Rate measure	Paired with Inactive Churn Rate on the Demographics page to test whether inactivity is a reliable leading indicator of churn. If the gap between the two rates is large, inactivity-triggered outreach is validated as a worthwhile investment.	Active Churn Rate = CALCULATE([Churn Rate], Dim_Customer[Is_Active] = "Yes")
Inactive Churn Rate	Tests the hypothesis that disengaged customers are more likely to formally churn.	A key early-warning metric. A significantly higher Inactive Churn Rate compared to Active Churn Rate validates an inactivity-triggered retention outreach programme as a worthwhile investment — catching customers before they formally exit.	Inactive Churn Rate = CALCULATE([Churn Rate], Dim_Customer[Is_Active] = "No")

DATA MODEL

The analytical model follows a **star schema** design, with the **churn_status** table serving as the central fact table and supporting dimension tables providing descriptive customer and calendar information. This structure minimizes data redundancy, improves report performance, and enables consistent calculations across all report pages.

Data Model Structure

Table	Type	Description
churn_status	Fact	Stores customer churn events and transactional attributes including churn status, churn reason, churn date, and customer identifier.
Dim_Customer	Dimension	Contains customer profile information such as age, region, account balance, tenure, product ownership, credit score, and activity status for segmentation and filtering.
Dates	Dimension	Provides a dedicated calendar table used for time intelligence calculations including Year-to-Date (YTD), Previous Year (PY), and moving averages.
_Metrics	Measure Table	Contains all DAX measures used throughout the report. This table improves model organization by separating calculated measures from raw data tables.

Relationship Design

The model uses **one-to-many relationships**, where each dimension table filters the central **churn_status** fact table.

- **Dim_Customer (1) → churn_status (*)**
  - Enables customer segmentation by demographics, geography, activity status, tenure, credit score, and account characteristics.
- **Dates (1) → churn_status (*)**
  - Supports trend analysis and time intelligence calculations such as Year-to-Date, Previous Year, and 3-month moving averages.

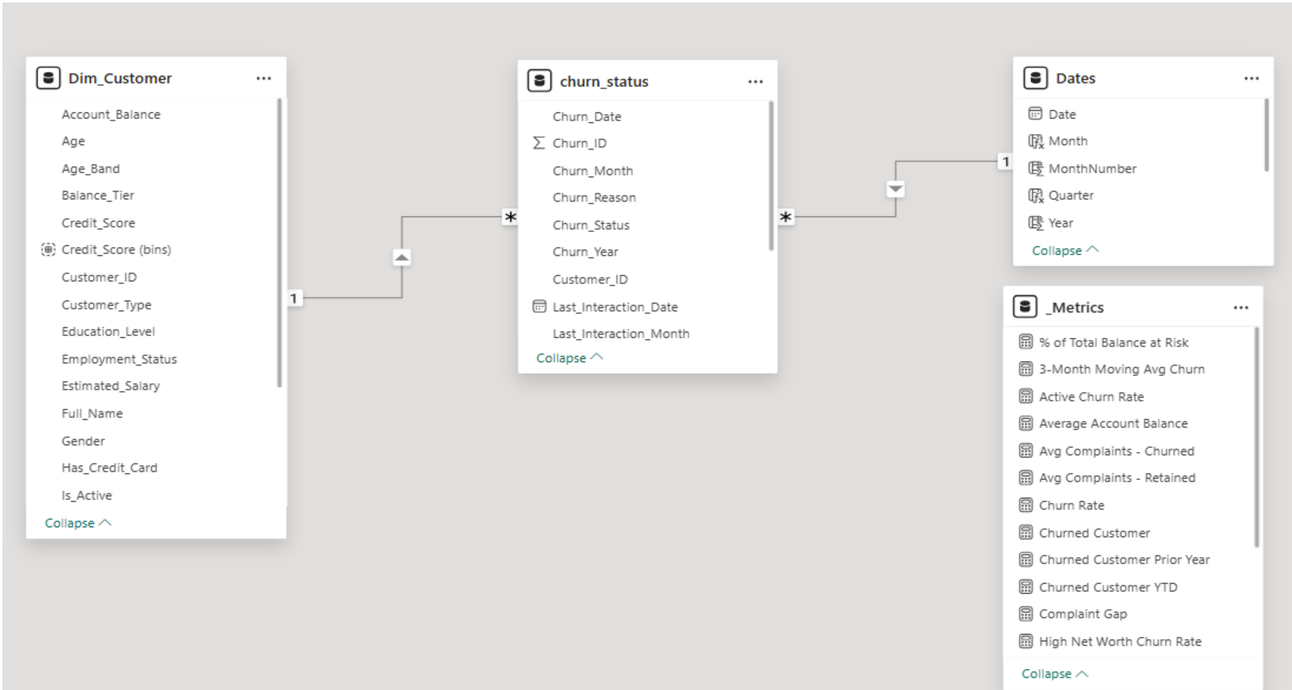

Figure 1. Star schema data model used for the FirstEdge Bank Customer Churn Analysis solution.

Conclusion

Customer attrition at FirstEdge Bank is primarily a competitive and engagement challenge rather than a service-quality problem. The analysis demonstrates that customer inactivity, limited product adoption, and regional competitive pressures provide measurable early indicators of churn. By implementing targeted retention initiatives supported by continuous monitoring through the accompanying Power BI dashboard, management can reduce customer attrition, protect deposit balances, and improve long-term customer lifetime value.